

Shallow water analysis

Yaoxuan Zeng

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Corrections in red.

Green: the cross-channel superelevation pressure $-g \partial_n h'$ (non-dimensional coefficient $\varepsilon_2/(\varepsilon_1 Fr^2 \alpha^2) \sim 1/(Fr^2 \alpha^2) \geq 1$) — the shallow-water term that can never be small.

1 Shallow water equation

We write the shallow water equation

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -g \nabla \eta - \frac{C_f |\mathbf{u}| \mathbf{u}}{h}, \quad (1)$$

$$\partial_t h + \nabla \cdot (\mathbf{u} h) = 0. \quad (2)$$

Let z_b be the bed elevation, so that $h = \eta - z_b$ and $\nabla \eta = \nabla h + \nabla z_b$. We write it in channel-following coordinate:

$$\partial_t u + u \partial_s u + v \partial_n u + C u v = -g \partial_s (h + z_b) - \frac{C_f |\mathbf{u}|}{h} u, \quad (3)$$

$$\partial_t v + u \partial_s v + v \partial_n v - C u^2 = -g \partial_n (h + z_b) - \frac{C_f |\mathbf{u}|}{h} v, \quad (4)$$

$$\partial_t h + \partial_s (u h) + \partial_n (v h) + C v h = 0. \quad (5)$$

Assume $u = U_0 + u'$, $v = v'$, $C = C'$, $h = H_0 + h'$, and assume that $u'/U_0 \sim h'/H_0 \sim \mathcal{O}(\varepsilon)$ ($z_b \sim H_0 = \mathcal{O}(1)$: the bed sits in the base state ($H_0 + z_b$), not the perturbation — only its secondary-flow tilt z'_b of §3 is $\mathcal{O}(\varepsilon)$). If we neglect all $\mathcal{O}(\varepsilon^2)$ terms, we have

$$\begin{aligned} \partial_t u' + U_0 \partial_s U_0 + U_0 \partial_s u' + u' \partial_s U_0 + v' \partial_n U_0 \\ = -g \partial_s (H_0 + z_b) - g \partial_s h' - \frac{C_f U_0^2}{H_0} - \frac{2C_f U_0 u'}{H_0} + \frac{C_f U_0^2}{H_0^2} h' \end{aligned} \quad (6)$$

$$\partial_t v' + U_0 \partial_s v' - C' U_0^2 = -g \partial_n (H_0 + z_b) - g \partial_n h' - \frac{C_f U_0 v'}{H_0} \quad (7)$$

$$\partial_t h' + \partial_s (U_0 H_0) + \partial_s (U_0 h') + \partial_s (u' H_0) + \partial_n (v' H_0) = 0 \quad (8)$$

Leading order balance:

$$U_0 \partial_s U_0 = -g \partial_s (H_0 + z_b) - \frac{C_f U_0^2}{H_0}, \quad (9)$$

$$-g \partial_n (H_0 + z_b) = 0, \quad (10)$$

$$\partial_s (U_0 H_0) = 0. \quad (11)$$

With $I \equiv -\partial_s z_b$ the valley slope, (9)–(11) are satisfied by $H_0(n) + z_b(n) = \text{const}$ together with $C_f U_0(n)^2 / H_0(n) = gI$, that is

$$U_0(n) = U_0 \tilde{\xi}(\tilde{n}), \quad H_0(n) = H_0 \tilde{\xi}(\tilde{n})^2, \quad C_f U_0^2 / H_0 = gI.$$

The depth profile is therefore slaved to the velocity profile: $H_0 \propto \tilde{\xi}^2$. A quadratic jet $\tilde{\xi} = 1 + r(1 - \tilde{n}^2)$ hence forces a *quartic* bed. The two profiles cannot be prescribed independently.

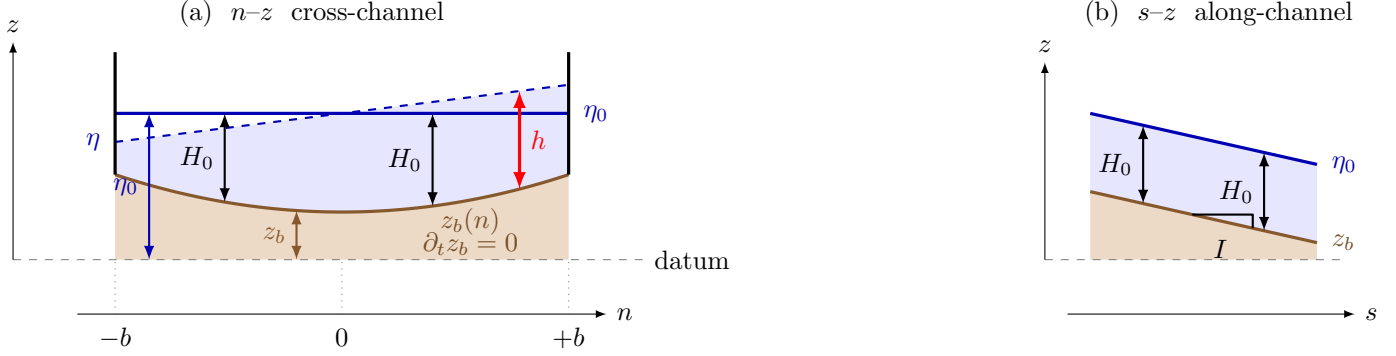


Figure 1: Elevations are measured from the datum; depths are the gaps between them. η_0 = base free-surface elevation, z_b = bed elevation, $H_0 = \eta_0 - z_b$ = base depth, $h = \eta - z_b = H_0 + h'$ = actual depth. (a) The bed is non-flat in n ; (10) makes η_0 flat across the section, so $H_0(n) = H_0 \tilde{\xi}(\tilde{n})^2$ varies with n through the bed alone. The dashed surface $\eta = \eta_0 + h'$ is the perturbation; the gap between the two blue lines is h' . (b) z_b and η_0 drop together at the valley slope $I = -\partial_s z_b$, and $-g\partial_s(h + z_b) = -g\partial_s \eta = gI$ drives the flow. Note $\partial_s z_b \neq 0$ — the bed elevation must fall downstream — while H_0 and the cross-sectional shape are the same at every s . “The bed does not change” means $\partial_t z_b = 0$, not $\partial_s z_b = 0$.

We retain the time dependence, and non-dimensionalize the $\mathcal{O}(\varepsilon)$ equation with

$$\tilde{u} = u'/U, \quad \tilde{v} = v'/V, \quad \tilde{s} = s/\Lambda, \quad \tilde{n} = n/b, \quad C' = C\tilde{C}(s), \quad \tilde{h} = h'/\mathcal{H}, \quad \tilde{t} = t/T, \quad T = \Lambda/U_0, \quad \Lambda = \lambda/2\pi, \quad (12)$$

we have

$$\frac{UU_0}{\Lambda} \tilde{\partial}_t \tilde{u} + \frac{UU_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{VU_0}{b} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{g\mathcal{H}}{\Lambda} \tilde{\partial}_s \tilde{h} - \frac{2C_f U_0 U}{H_0} \frac{\tilde{u}}{\tilde{\xi}} + \frac{C_f U_0^2 \mathcal{H}}{H_0^2} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (13)$$

$$\frac{VU_0}{\Lambda} \tilde{\partial}_t \tilde{v} + \frac{VU_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{v} - C U_0^2 \tilde{C} \tilde{\xi}^2 = -\frac{g\mathcal{H}}{b} \tilde{\partial}_n \tilde{h} - \frac{C_f U_0 V}{H_0} \frac{\tilde{v}}{\tilde{\xi}}, \quad (14)$$

$$\frac{\mathcal{H}U_0}{\Lambda} \tilde{\partial}_t \tilde{h} + \frac{\mathcal{H}U_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{h} + \frac{H_0 U}{\Lambda} \tilde{\partial}_s (\tilde{\xi}^2 \tilde{u}) + \frac{H_0 V}{b} \tilde{\partial}_n (\tilde{\xi}^2 \tilde{v}) = 0. \quad (15)$$

Define

$$\alpha = b/\Lambda, \quad \gamma = V/U, \quad \varepsilon_1 = U/U_0, \quad \varepsilon_2 = \mathcal{H}/H_0, \quad Fr = U_0/\sqrt{gH_0}, \quad F_c = C_f \Lambda/H_0, \quad Ci = Cb \quad (16)$$

we have

$$\tilde{\partial}_t \tilde{u} + \tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{\gamma}{\alpha} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \frac{\tilde{u}}{\tilde{\xi}} + F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (17)$$

$$\tilde{\partial}_t \tilde{v} + \tilde{\xi} \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha \gamma} \tilde{C} \tilde{\xi}^2 = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha \gamma} \tilde{\partial}_n \tilde{h} - F_c \frac{\tilde{v}}{\tilde{\xi}}, \quad (18)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \frac{\gamma}{\alpha} \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} (\tilde{\partial}_t \tilde{h} + \tilde{\xi} \tilde{\partial}_s \tilde{h}) = 0. \quad (19)$$

The green term is the cross-channel super-elevation pressure: the free surface tilts across the channel to balance the centrifugal forcing $\propto Ci \tilde{C}$, and its streamwise gradient sets the near-bank u'_b (Ikeda's $-U_0^2 \partial_s C'$). Its coefficient $\varepsilon_2/(\varepsilon_1 Fr^2 \alpha \gamma) \sim 1/(Fr^2 \alpha^2)$ is ≥ 1 for every subcritical ($Fr < 1$), long-wave ($\alpha < 1$) meander — $O(1/\varepsilon)$ in limit 1, $O(1)$ in limit 2 — so it can never be small. That is why limit 2 (the QGPV limit, which *keeps* it) still migrates downstream. Discarding the free surface entirely (rigid lid, no $\tilde{h}$) is the only way to remove it; the pure-vorticity model that then remains migrates upstream on its own β -Rossby dynamics.

We typically assume $\alpha = \gamma$. If we assume $Ci \sim \varepsilon_1 \sim \varepsilon_2 \sim \mathcal{O}(\varepsilon)$, $\alpha \sim \mathcal{O}(\varepsilon^{1/2})$, $Fr \sim F_c \sim 1$, the leading order equation becomes

$$\tilde{\partial}_t \tilde{u} + \tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \frac{\tilde{u}}{\tilde{\xi}} + F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (20)$$

$$\frac{1}{\varepsilon_1 \alpha^2} Ci \tilde{C} \tilde{\xi}^2 = \frac{1}{\varepsilon_1 \alpha^2} \frac{\varepsilon_2}{Fr^2} \tilde{\partial}_n \tilde{h}, \quad (21)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} (\tilde{\partial}_t \tilde{h} + \tilde{\xi} \tilde{\partial}_s \tilde{h}) = 0. \quad (22)$$

Equations 20 and Equation 21 lead to the solution in Ikeda et al.

If we assume that $Ci \sim \varepsilon_2 \sim \mathcal{O}(\varepsilon^2)$, $\varepsilon_1 \sim Fr^2 \sim \mathcal{O}(\varepsilon)$, and set $\alpha = 1$, the leading order equation becomes

$$\tilde{\partial}_t \tilde{u} + \tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \frac{\tilde{u}}{\tilde{\xi}}, \quad (23)$$

$$\tilde{\partial}_t \tilde{v} + \tilde{\xi} \tilde{\partial}_s \tilde{v} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_n \tilde{h} - F_c \frac{\tilde{v}}{\tilde{\xi}}, \quad (24)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) = 0. \quad (25)$$

Equations 23-25 leads to the QGPV equation

$$\tilde{\partial}_t \tilde{\zeta} + \tilde{\xi} \tilde{\partial}_s \tilde{\zeta} + \tilde{\beta} \tilde{v} = -\frac{F_c}{\tilde{\xi}} \tilde{\zeta} + \frac{F_c}{\tilde{\xi}} \tilde{\partial}_n \tilde{u} - \frac{2F_c \tilde{u}}{\tilde{\xi}^2} \tilde{\partial}_n \tilde{\xi}, \quad (26)$$

where

$$\tilde{\beta} = 2 \frac{(\tilde{\partial}_n \tilde{\xi})^2}{\tilde{\xi}} - \tilde{\partial}_{nn} \tilde{\xi}, \quad \tilde{\zeta} = \tilde{\partial}_s \tilde{v} - \tilde{\partial}_n \tilde{u}, \quad \tilde{\xi}^2 \tilde{u} = -\tilde{\partial}_n \tilde{\psi}, \quad \tilde{\xi}^2 \tilde{v} = \tilde{\partial}_s \tilde{\psi}. \quad (27)$$

We can put:

$$\tilde{\partial}_t \tilde{u} + \tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi}(\tilde{n}) + \frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} + 2F_c \frac{\tilde{u}}{\tilde{\xi}(\tilde{n})} - F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}(\tilde{n})^2} = 0, \quad (28)$$

$$\tilde{\partial}_t \tilde{v} + \tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C}(\tilde{s}) \tilde{\xi}(\tilde{n})^2 + \frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha^2} \tilde{\partial}_n \tilde{h} + F_c \frac{\tilde{v}}{\tilde{\xi}(\tilde{n})} = 0, \quad (29)$$

$$\tilde{\partial}_s(\tilde{\xi}(\tilde{n})^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}(\tilde{n})^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} (\tilde{\partial}_t \tilde{h} + \tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{h}) = 0. \quad (30)$$

2 Bank erosion

Let $y_b(s, t)$ be the lateral position of a bank and $u'_b = u'(s, \pm b)$ the near-bank velocity perturbation. With $\psi'_1, \psi'_2, \psi'_3$ the streamfunction perturbation at $n = +b, 0, -b$,

$$\psi'_1 = \bar{u}(b) y_b = U_0 y_b, \quad (31)$$

$$u'_b = (\psi'_2 - \psi'_1)/b, \quad (32)$$

$$\partial_t \psi'_1 = \frac{\varepsilon C_f U_0}{b} (\psi'_2 - \psi'_1), \quad (33)$$

$$\gamma \partial_t y_b = E u'_b, \quad E = \varepsilon C_f, \quad \gamma = \cos \theta. \quad (34)$$

Equation (33) is the three-level form; (34) is Ikeda, Parker & Sawai (1981) eqs (11)–(13). For two independently mobile banks y_N, y_S ,

$$\partial_t y_N = +E_{e,d} u'_N, \quad \partial_t y_S = -E_{e,d} u'_S, \quad (35)$$

with E_e where $u' > 0$ (erosion) and E_d where $u' < 0$ (deposition); $E_e = E_d$ gives constant width.

3 Secondary flow and the bed — the A switch

The note has no secondary-flow term; a depth-averaged 2-D model cannot generate the helical circulation of a bend, so its effect on the bed is *imposed* as a curvature-slaved perturbation to the frozen depth of §1. Writing the total depth as $H(s, n) = H_0(n) + H'_2(s, n)$ with $H_0(n)$ the time-independent base profile, Ikeda, Parker & Sawai (1981) eq. (6) is

$$H'_2(s, n) = -A H_0(n) \mathcal{C}'(s) n, \quad n = b\tilde{n}, \quad (36)$$

a pool ($H'_2 > 0$, deeper) on the outer bank and a point bar on the inner bank, both proportional to the local centreline curvature $\mathcal{C}'(s)$. (This is Ikeda's bed perturbation η' , renamed to avoid clashing with the free surface η ; the sign is fixed so the pool is on the outer bank. It enters the bed of §1 as $z_b = \bar{z}_b(n) + z'_b$ with $z'_b = -H'_2$.) The secondary-flow strength is a switch,

$$A = 0 : \text{ incised, no secondary flow } (H'_2 \equiv 0, \text{ frozen bed}); \quad A = 2.89 : \text{ alluvial (Suga 1963)}. \quad (37)$$

$H_0(n)$ never changes in time; H'_2 carries no independent dynamics — it is slaved to the slowly migrating planform (recomputed from the current curvature each morphological step), so “the bed does not change” still means $\partial_t z_b = 0$. With $h' = \xi' - \eta'$ and Ikeda's superelevation $\xi' = \mathcal{C}' U_0^2 n / g$, the depth perturbation is $h' / H_0 = (F^2 + A) \mathcal{C}' \tilde{n}$: A and F^2 occupy the *same* depth/drag slot, so $A = 0$ leaves only $F^2 \approx 0.09$ while $A = 2.89$ raises the bend forcing to ≈ 2.98 — which is what turns bend decay into growth.

4 Boundary conditions

The note is an interior linear analysis and states no boundary conditions; the model closes the channel $[0, L] \times [-b, b]$ with

$$\underbrace{u_n = -\bar{u}(n)}_{\text{inflow } s=0}, \quad \underbrace{\eta = \eta_{\text{ref}} - I L}_{\text{outflow } s=L}, \quad \underbrace{u_n = 0}_{\text{banks } n=\pm b}. \quad (38)$$

The flow is subcritical ($Fr < 1$), so exactly one characteristic enters at each open boundary: the inflow prescribes the velocity profile only, the outflow the elevation only. Pinning η , u and v together at one open boundary would over-determine it (and, started from rest, launch a spurious bore). The banks are free-slip — no normal flux, no wall drag beyond the bed friction. Drag is quadratic, $-C_f |\mathbf{u}| \mathbf{u} / h$ with $C_f = 0.05$; a small horizontal viscosity $\nu = 0.05 \text{ m}^2 \text{ s}^{-1}$ is retained for DG stability (both stated here because the note carries neither).